TECHNICAL REPORT ON STUDENTS INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES)

AT
LOKUWA PRIMARY HEALTHCARE CENTRE,
MUBI NORTH, ADAMAWA STATE

FROM: 7TH OCTOBER, 2024
TO: 25TH JANUARY, 2025

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ST/SLT/ND/23/157

SUBMITTED TO THE DEPARTMENT OF SCIENCE LABORATORY TECHNOLOGY, SCHOOL OF SCIENCE AND TECHNOLOGY, FEDERAL POLYTECHNIC, MUBI, ADAMAWA STATE.

IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF NATIONAL DIPLOMA (ND) IN SCIENCE LABORATORY TECHNOLOGY,

JUNE, 2025

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Definition of SIWES

The Student Industrial Work Experience Scheme (SIWES) was designed by the Federal Government Establishment by decree No. 47 with statuary responsibility of man power training to bridge the gap between theoretical class work and the actual practical experience which is really what the labor market require of a graduate.

The program was established in 1973 and board as industrial Training Fund (ITF) with the view of generating trained manpower sufficient to meet the needs of the country's development plan. Despite the obstacles and huddles such as accommodation problems during the program, these are worthy adventure. I salute the pioneers and right-thinking person that initiate (SIWES) (Ramayah & Lee, 2012).

Purpose of SIWES

In the earlier stage, students are graduating without any technical knowledge or working experience and this makes them to undergo further training after securing an employment. With this reason, student industrial training was established. During this program, as designed by the ITF, students are expected to get technical assistance and acquire more experience scheme in their chosen field of study and exposed them to the usage of source machines and safety precaution where relevant before the completion of their program in their various institutions.

Objectives of SIWES

The objectives of SIWES is to bridge the gap between theories learned and the practical aspect found in the field or industry as it may be. In addition, the following are some of the aim and objective.

i. To strengthen the relationship between institution of learning and the industries.
ii. To expose and prepare student for the industrial challenges that is likely to be encountered in the future.
iii. To create the mind set of industrial based skills necessary for smooth transition from class room to the world of application (field) after graduation.
iv. To expose students to know how to handle, maintain and make use of tools and equipment that may not be available at their institutions of learning.
v. The SIWES also aims at preparing student for work after graduation (Ramayah & Lee, 2012).

METHOD OF BLOOD SAMPLE COLLECTION

There are two ways medically through which blood sample can be collected. The ways are as follows:

a. Through vein
b. Through capillary

Vein Blood Sample Collection

Materials required: Swap, EDTA container, Tunicate, Syringe and Needle.

Procedure:

i. The patient hand was tied with a tunicate
ii. The preferred area of the patient was wiped using a swap, so that the vein will be seen clearly.
iii. The patient was introduced to close his tied hand fingers so that the vein will pump up.
iv. Then about 1/3 of the needle was inserted into the vein and the plump of the syringe was drawn back for the blood sample to be collected.
v. After the blood sample was collected, the tunicate was untied first before the needle was removed from the vein.
vi. Then the blood sample was transferred into the EDTA container.

Capillary Blood Sample Collection

Materials required: Swap, EDTA container, Lancet.

Procedure:

i. The patient's finger was wiped using a swap.

ii. The wiped area was pricked with a lancet.

iii. The finger was pressed gently for free flow of blood.

iv. Then the blood sample was collected into the EDTA container.

Parasitology is the branch of biology or medicine concern with the study of parasitic organism (Hoffman *et al.*, 2018). Tests done under this unit include:

Malaria Parasite (MP)

Specimen: Blood

Reagent: Buffer Solution

Materials: Malaria kit, wire loop, anti-coagulant container syringe, tourniquet, spirit swap.

Procedure:

i. The tourniquet was put above the area to get the veins to swell with blood.

ii. Inserted a needle into a vein (usually in the arm inside of inside of the elbow or on the back of the hand)

iii. The tourniquet was taken off and the needle removed the needle from the vein.

iv. The wire loop was used to pick some blood sample from the anticoagulant container.

v. Inserted it in to sample space.

vi. Added two to four drops of buffer in the space provided for it.

vii. Observed and saw the result after 15 to 20 minutes

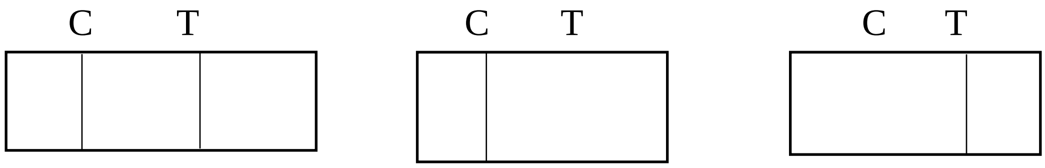
Result

i. If there is a line on the test column and on the control column the result is positive for malarial parasites.

ii. And if there is only one line on the control column and non on the test column it is negative for malaria Parasite.

iii. If there is a line on the column and non on the control column it is invalid.

iv. If there is no line on the control and test column is still invalid.



Urine analysis

It is used to detect and manage a wide range of disorders, such as urinary tract infections, kidney disease and diabetes.

Aim: To detect the normality or abnormality of urine.

Urine Analysis Test Procedure:

i. The urine was collected in a clean container.

ii. The urine sample was poured on a test strip and allowed for 60 seconds.

iii. The colour chart on a test strip was compared with the colour chart provision on the container of the test strip before the result was interpreted.

Result:

i. The observation of the result depends on the colour change on the strip if any.

ii. Then the strip was compared with the colour of the parameter of the combi 9 strip container.

HAEMATOLOGY UNIT

Haematology is the study of blood in its normal and abnormal condition and the tests carried out under this unit include (Hoffman *et al.*, 2018).

A B O Grouping

ABO Grouping

Aim: To help identify the blood group of an individual

Specimen: Blood

Reagent: ABO grouping sera (anti A, B, and ORh)

Materials: White surface tile, Pasteur pipette, stirrer, toilet tissue, syringe tile, Pasteur pipette, stirrer, toilet tissue, syringe, tourniquet, swap.

Procedure:

i. Cleaned the skin with swap

i. Put a tourniquet above the area to get the veins to swell with blood.

ii. Inserted a needle into a vein either on the back of the hand or in the arm inside of the elbow.

iii. Pulled the blood sample into the syringe

iv. Took off the tourniquet and remove the needle from the vein and pour the blood on the white surface tile in different places.

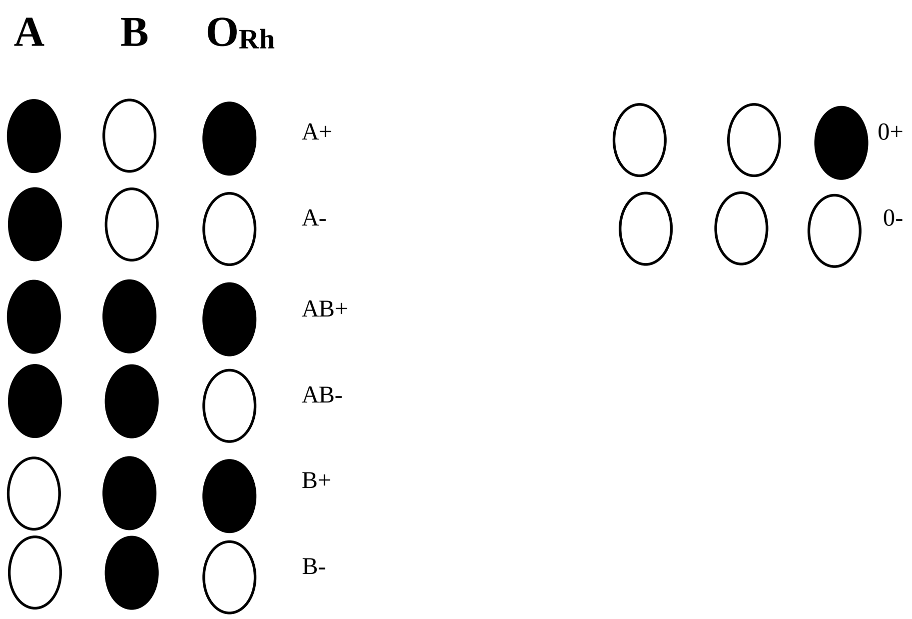
v. Added anti A, anti B and Orh on each drop of blood on the white surface tile.

vi. Used the stirrer to mix both reagent and blood sample properly.

vii. Rock from 0 -1 minute and read your result.

Result

i. If anti-A agglutinate, and no agglutination in anti B and ORh agglutinate it is A positive.
ii. If there is agglutination in anti A and non in anti B and ORh it is A negative.
iii. If there is agglutination in anti A, and anti B and ORh it is Ab positive.
iv. If there is no agglutination in anti A and there is in anti B and ORh it is B positive.
v. If there is no agglutination in anti A and there is in anti B but non in ORh it is B negative.
vi. If there is no agglutination in anti A, anti B but there is in ORh it is O positive.
vii. If there is no agglutination in anti A, anti B and ORh it is O negative.



IMMUNOLOGY UNIT

This is the study of human immunity and tests done under this unit include

Pregnancy test (PT)

Retro viral screening or syndrome (RVS)

Pregnancy Test (PT)

Aim: To delete HCG in a female blood or urine

Specimen: blood serum or early morning urine

Material: pregnancy test strip, test tube

Procedure:

i. unveiled the pregnancy strip

ii. Inserted the PT strip into the blood serum or early morning urine in the test tube.

iii. Made sure the strip stick is covered to the marked point by either of the sample used.

iv. Allowed to stand for some few minutes and then remove the strip stick from the sample and observe.

Result

a. If there are two lines at both the control and test column it is positive for pregnancy test.
b. If there is only one line at the control column and non at the test column is negative for pregnancy test.
c. If there is one line at the test column and non at the control column the result is invalid.
d. If there are no lines at both the control and the test column is still invalid.

Retro Viral Syndrome/Screening (RVS)

Aim: To detect immune virus in the blood

Specimen: Blood Serum

Reagent: Buffer Solution

Materials: RVS strip, Pasteur pipette

Procedure

i. Unveiled the RVS strip
ii. Used Pasteur pipette to make one or two drops of serum on the space provided
iii. Added buffer, to re-dilute the serum

Result

i. If there are two lines at both control and test column it is positive for RVS test
ii. If there is only one line at the control column and non at the test column it is negative for RVS test.
iii. If there is only one line at the test column and non at the control column the result is invalid.

CHEMICAL PATHOLOGY UNIT

This is the study of chemicals and test carried out under this unit include (Hoffman *et al.*, 2018). Tests carried out under this unit include: Widal Test, H-pylori

Widal Test

Aim: To diagnose typhoid in the blood

Specimen: Blood serum

Reagent: Salmonella antigen sera

Apparatus: White surface tile, Pasteur pipette, stirrer, tourniquet, syringe, spirit swap.

Procedure

i. Cleaned the skin with spirit swap

ii. Put a tourniquet above the area to get the veins to swell with blood

iii. Inserted the needle into a vein either in the arm inside of the elbow or in the hand

iv. Pulled the blood sample into the syringe

v. Took off the tourniquet and remove the needle from the vein

vi. Poured the blood into an anti-coagulant container.

vii. Allowed the blood to stand for some time in order to get the serum.

viii. Eight drops of serum were put on a white surface tile using your Pasteur pipette.

ix. Added salmonella antigen sera drops in each of the eight of the serum on the tile.

x. Used the stirrer to mix both the serum and the reagent properly and rock from 0 - 1 minute.

Result

i. Observe if there is thick agglutination the result is positive for typhoid fever titre 1 in 320.

ii. If the agglutination is fairly thick it is still positive for typhoid fever titre 1 in 160.

iii. But if there is no agglutination it is negative for typhoid fever the titre could be 1 in 120, 1 in 40, and 1 in 80.

Helicobacter pylori (H. pylori) Test

Helicobacter pylori (H. pylori) is a type of bacteria that infects the digestive system.

Aim: To Look for H. pylori bacteria in the digestive tract

H-Pylori Test Procedure:

i. The patient's urine was collected in a sterile test tube.

ii. The urine sample was then placed into the centrifuge machine. The machine was allowed to run for 5 minutes, after which the centrifuged sample was carefully removed.

iii. Taking a clean slide, the centrifuged urine sample was placed onto it.

iv. The slide was then observed under a microscope, using the x100 objective lens.

Result

i. The observation of the result depended on any visible changes in color on the strip that was used during the test.

ii. The strip was then compared with the color parameters provided on the combi 9 strip container to interpret the results accurately.

Human Immune Virus (HIV)

Aim: To detect the presence of Human Immunodeficiency Virus (HIV) in the blood.

Specimen: Whole blood (capillary blood from fingertip).

Reagent: HIV test strip, buffer solution

Apparatus: Test strip (rapid diagnostic test kit), buffer solution, lancet, alcohol swab, timer

Procedure

i. Cleaned the fingertip of the patient using an alcohol swab and allowed it to dry.
ii. Pricked the finger with a sterile lancet.
iii. Collected a small drop of blood and placed it on the sample area of the HIV test strip.
iv. Added the buffer solution onto the same area of the test strip as instructed by the kit manufacturer.
v. Allowed the test strip to stand undisturbed for 15 minutes.
vi. Interpreted the result as follows:

Result

i. If two lines appear (one on the control line and one on the test line), the result is **positive** for HIV.
ii. If only one line appears on the control line, the result is **negative** for HIV.
iii. If no line appears or only the test line shows without the control line, the result is **invalid** and the test should be repeated.

Genotype Test

Aim: To determine the haemoglobin genotype of an individual (e.g., AA, AS, SS, AC).

Specimen: Whole blood (preferably venous blood)

Reagent: Hemoglobin electrophoresis buffer, cellulose acetate paper, staining solution

Apparatus: Electrophoresis chamber, cellulose acetate strip, power supply, test tubes, micropipette, centrifuge, staining tray

Procedure

i. Cleaned the skin at the venipuncture site using a spirit swab.
ii. Applied a tourniquet above the site to locate a suitable vein.
iii. Inserted a sterile needle into the vein and collected blood using a syringe.
iv. Transferred the blood into an EDTA (anti-coagulant) tube.
v. Centrifuged the blood sample to separate red blood cells (RBCs) from plasma.
vi. Prepared a haemolysate from the RBCs by mixing with a haemolysing agent.
vii. Spotted the haemolysate on a cellulose acetate strip.
viii. Placed the strip in the electrophoresis chamber containing the buffer solution.
ix. Connected the power supply and ran electrophoresis for the required time.
x. After completion, removed and stained the strip to observe band positions.

Result

i. If a single band is seen in the position of HbA, the genotype is **AA**.
ii. If two bands are seen in the positions of HbA and HbS, the genotype is **AS**.
iii. If a single band is seen in the position of HbS, the genotype is **SS**.
iv. If bands are seen for HbA and HbC, the genotype is **AC**.
v. Other combinations may indicate rare or abnormal genotypes and should be further confirmed.

Problems Encountered

i The time frame set for the program is too short as some of the aspects of the program where not completed.

ii Lack of Financial support from the company to aid transportation to and from training.

iii Attentions are not given to the IT students by the workers it is learn if you want to learn or ask if you want to know.

Recommendations

This student industrial work experience scheme (SIWES) programme was interesting; some students find it difficult to carry out some of the test with modern equipment in the laboratory. Likewise, there was restriction on some of the equipment which is only operated by the specialist in the establishment. In this respect I am pleading with the government to provide modern equipment to school. The laboratory and the student should be given opportunity to operate the available equipment in school laboratory (Physics, Chemistry, and Microbiology).

Secondly the ITF official should try to visit establishment where students are undergoing their SIWES program.

Finally, my advice to incoming student who may undergo their course of study and the establishment should be equipped with modern equipment.

Summary

This Student Industrial Work Experience Scheme program has really enlightened us a lot of things that we have not done during our theoretical study and it also exposed us to many principles, test and some machines that we have not seen or handled before.

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